

Mathematical modeling of titration curves

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It has frequently been observed that the shapes of titration curves closely resemble those of inverse hyperbolic or trigonometric functions. Here we show that there exists not only a resemblance, but that, for simple titration reactions, we can derive expressions which represent the change in a solution property with the volume of titrant added, in terms of inverse hyperbolic or trigonometric functions. The general mathematical form of the expressions so obtained is then used to model these and other, more complicated titrations. Copyright © 2002 John Wiley & Sons, Ltd.

KEYWORDS: volumetric titrations; mathematical modeling; titration curves; Gran plots

1. INTRODUCTION

Volumetric titrations are standard techniques that use a change in a specific property of the solution to follow the changes in the solution concentration when a titrant is added. The different types of titration, i.e. neutralization, precipitation, complex formation, redox, etc., are assigned as standard undergraduate experiments. On the other hand, the analysis of the titration process to determine the equivalence point usually involves the solution of higher-order polynomial equations if an exact quantitative analysis is required. Most textbooks announce from the beginning that they will not give consideration to equations of degree higher than the second. To avoid the use of higher-order equations, the authors introduce some assumptions that lead to simpler equations. This is usually done by dividing the titration curve into several zones where simple equations are valid. However, this kind of approach has several shortcomings. Students could be led to believe that a complete, exact treatment of the problem cannot be performed and are induced to see titrations as discontinuous processes. Also, this kind of approach introduces numerical errors, since the range of validity of a certain assumption is not clear, especially at the boundaries between zones.

Our aim here is to show that the titration curve has a form that allows it to be modeled by certain mathematical functions, already learned by the beginning chemistry student in high school or in a beginning undergraduate math course. To our knowledge, the relations have not been presented previously in the chemical literature, although some authors have noted a similarity in the shapes of the titration curves and these kinds of functions [1]. We show that they are not merely similar, but that, as our mathematical analysis demonstrates, these functions emerge naturally

from the analysis of the titration equations. The theoretical analysis allows us to suggest a general function that can be used to fit data from more complicated titrations and to determine with accuracy the equivalence point. Our results will be illustrated with a few examples which include acid–base neutralization, precipitation, complexometric and redox titrations. Further, we perform some linearization procedures on our non-linear expressions to show how Gran and Gran-type equations [2] emerge as approximations to the equations derived in this paper. Finally, we point out a remarkable similarity between the curve of titration and the integrated absorption curve in NMR, which allows us to draw an analogy between the absorption of radiation by a nucleus and the neutralization of a given volume of an analyte with titrant volume.

2. SOME REMARKABLE MATHEMATICAL RELATIONS

There exist some very interesting relations between the inverse hyperbolic or trigonometric functions and the natural logarithm function. The formulae presented here will be used to express titration parameters such as pH, pM (M being the molar concentration of any species being titrated) and cell voltage as functions of titrant volume.

The expressions derived here make use of the following remarkable relations between inverse hyperbolic functions and the natural logarithm, which we give here without proof [3, 4]:

$$\operatorname{arcsinh}(x) = \ln\left(x + \sqrt{1+x^2}\right) \quad (1)$$

$$= \operatorname{arccosh}\left(\sqrt{x^2+1}\right) \quad (2)$$

$$= \operatorname{arctanh}\left(\frac{x}{\sqrt{x^2+1}}\right) \quad (3)$$

There exist equivalent relations for inverse trigonometric functions which involve complex numbers.

It has often been observed that titration curves look like

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those of inverse trigonometric functions (or inverse hyperbolic functions, since they are related through a transformation). However, to our knowledge, no theoretical explanation has been given for that apparent connection.

Suppose that we can describe the change in a certain property x of the solution being titrated (e.g. the molar concentration of H_3O^+) by a second-degree polynomial equation

$$x^2 + bx + c = 0, \quad b, c < 0 \quad (4)$$

where the parameters b and c are, in general, dependent on the titrant volume and other parameters of the solution.

Choosing the positive solution of Equation (4), we find

$$\begin{aligned} x &= -\frac{b}{2} + \frac{1}{2}\sqrt{b^2 - 4c} \\ &= -\frac{b}{2} + \sqrt{\frac{b^2}{4} - c} \end{aligned} \quad (5)$$

We now express Equation (5) in a dimensionless form by multiplying it by a parameter β :

$$x\beta = -\frac{b\beta}{2} + \sqrt{\frac{(b\beta)^2}{4} - c\beta^2} \quad (6)$$

Letting

$$x\beta = \zeta + \sqrt{\zeta^2 + 1} \quad (7)$$

ζ can be expressed in terms of b , c and β as

$$\zeta = \frac{\beta^2 \left(-\frac{b}{2} + \sqrt{\frac{b^2}{4} - c} \right)^2 - 1}{2\beta \left(-\frac{b}{2} + \sqrt{\frac{b^2}{4} - c} \right)} \quad (8)$$

Thus, taking logarithms in Equation (7) and using the relations given in Equations (1)–(3), we obtain

$$\text{px} = -\text{p}\beta - \frac{1}{\ln(10)} \text{arcsinh}(\zeta) \quad (9)$$

$$= -\text{p}\beta - \frac{1}{\ln(10)} \text{arccosh}(\zeta^2 + 1) \quad (10)$$

$$= -\text{p}\beta - \frac{1}{\ln(10)} \text{arctanh}\left(\frac{\zeta}{\sqrt{1 + \zeta^2}}\right) \quad (11)$$

where we have used the standard definition of the operator $\text{p} = -\log$. The value of β will be chosen such that at the equivalence point $\zeta = 0$ and, consequently, $\text{px} = -\text{p}\beta$ (since $\text{arcsinh}(0) = \text{arccosh}(1) = \text{arctanh}(0) = 0$).

In the next sections we study the simplest general problem of each of most known types of titrations to derive, with the help of this section, mathematical expressions which explain the observed similarity between titration curves and inverse hyperbolic or trigonometric functions.

3. NEUTRALIZATION TITRATIONS

3.1. Titration of a strong acid with a strong base

Consider the case of the titration of a volume V_a of strong acid of molar concentration M_a with a volume V_b of a strong

base of molar concentration M_b . In this case the concentration of H_3O^+ , $[\text{H}_3\text{O}^+]$, as a function of the volume V_b of titrant can be obtained by combining the mass and charge balance equations [5]. It is given by the second-degree polynomial equation

$$[\text{H}_3\text{O}^+]^2 - \frac{M_a V_a - M_b V_b}{V_b + V_a} [\text{H}_3\text{O}^+] - K_w = 0 \quad (12)$$

Equation (12) has the same form as Equation (4); thus, from Equation (8), we find

$$\begin{aligned} \zeta &= \frac{\beta M_a V_a - M_b V_b}{2(V_b + V_a)} \\ &+ \frac{\beta^2 K_w - 1}{2\beta \left(\frac{M_a V_a - M_b V_b}{2(V_b + V_a)} + \sqrt{\frac{(M_a V_a - M_b V_b)^2}{4(V_b + V_a)^2} + K_w} \right)} \end{aligned} \quad (13)$$

To find β , we impose the condition already stated that $\zeta = 0$ at the equivalence point and obtain, from Equation (13), $\beta = 1/\sqrt{K_w}$. Thus

$$\zeta = \frac{1}{2\sqrt{K_w}} \frac{M_a V_a - M_b V_b}{V_b + V_a} \quad (14)$$

In this way, using Equations (9)–(11), we can write the following equivalent expressions for the pH as a function of volume V_b and other properties of the solution:

$$\text{pH} = \frac{1}{2} \text{p}K_w - \frac{\text{arcsinh}\left(\frac{1}{2\sqrt{K_w}} \frac{M_a V_a - M_b V_b}{V_b + V_a}\right)}{\ln(10)} \quad (15)$$

$$= \frac{1}{2} \text{p}K_w - \frac{\text{arccosh}\left(\sqrt{\frac{1}{4K_w} \left(\frac{M_a V_a - M_b V_b}{V_b + V_a}\right)^2 + 1}\right)}{\ln(10)} \quad (16)$$

$$= \frac{1}{2} \text{p}K_w - \frac{\text{arctanh}\left(\frac{\frac{M_a V_a - M_b V_b}{V_b + V_a}}{\sqrt{\left(\frac{M_a V_a - M_b V_b}{V_b + V_a}\right)^2 + 4K_w}}\right)}{\ln(10)} \quad (17)$$

Any one of the preceding expressions will yield the same curve when plotting pH versus V_b , the volume (ml) of titrant.

The first and second derivatives of the function pH (V_b) are given by

$$\frac{\text{d} \text{pH}}{\text{d} V_b} = \frac{(M_a + M_b) V_a}{(V_a + V_b)^2 \sqrt{4K_w + \frac{(M_a V_a - M_b V_b)^2}{(V_a + V_b)^2}} \ln(10)} \quad (18)$$

and

$$\begin{aligned} \frac{\text{d}^2 \text{pH}}{\text{d} V_b^2} &= -\frac{(M_a + M_b)^2 V_a^2 (M_b V_b - M_a V_a)}{(V_a + V_b)^5 \left(4K_w + \frac{(M_a V_a - M_b V_b)^2}{(V_a + V_b)^2} \right)^{3/2} \ln(10)} \\ &- \frac{2(M_a + M_b) V_a}{(V_a + V_b)^3 \left(4K_w + \frac{(M_a V_a - M_b V_b)^2}{(V_a + V_b)^2} \right)^{1/2} \ln(10)} \end{aligned} \quad (19)$$

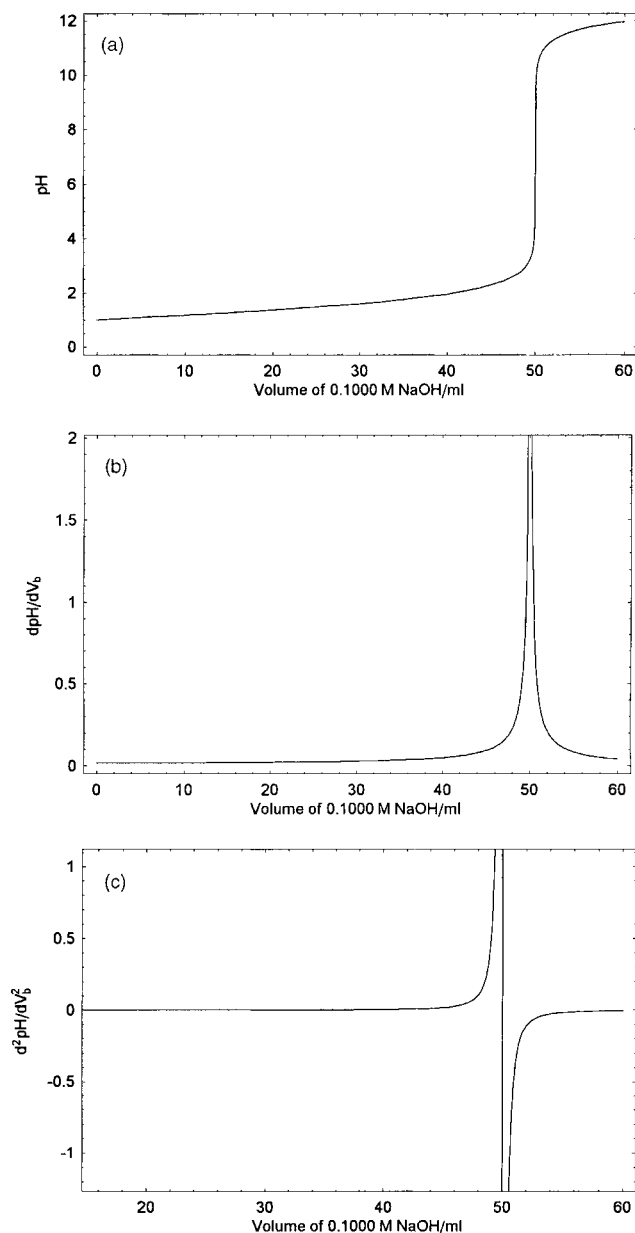


Figure 1. Titration of 50.00 ml of 0.1000 M HCl with 0.1000 M NaOH: (a) pH vs V_b ; (b) dpH/dV_b ; (c) d^2pH/dV_b^2 .

Our results, in this way, justify theoretically the use of relations which involve inverse hyperbolic or trigonometric functions for fitting strong acid–strong base titration experimental data. Further, if we use the relation between the arctan and the arctanh, Equation (17), for instance, can be rewritten as

$$pH = \frac{1}{2}pK_w + i \frac{\operatorname{arctanh}\left(i \frac{\frac{M_a V_a - M_b V_b}{V_b + V_a}}{\sqrt{\left(\frac{M_a V_a - M_b V_b}{V_b + V_a}\right)^2 + 4K_w}}\right)}{\ln(10)} + i \frac{\operatorname{arctanh}\left(i \frac{\frac{M_b V_b - M_a V_a}{V_a + V_b}}{\sqrt{\left(\frac{M_b V_b - M_a V_a}{V_a + V_b}\right)^2 + 4K_w}}\right)}{\ln(10)} \quad (20)$$

In spite of the presence of $i = \sqrt{-1}$, the preceding expression is real, since the original relation in terms of arctanh is real.

Figure 1(a) shows the titration curve obtained using Equations (15)–(17) for the case of the titration of 50.00 ml of a 0.1000 M strong acid with 0.1000 M strong base. Also shown in Figures 1(b) and 1(c) are the first and second derivatives of pH versus V_b obtained with Equations (18) and (19). As observed, our expressions reproduce the common textbook titration curves and their derivatives. Further, Equations (15)–(17) reveal that at the equivalence point, where $V_b = M_a V_a / M_b$, $pH = \frac{1}{2}pK_w = 7$, since $\operatorname{arcsinh}(0) = \operatorname{arccosh}(1) = \operatorname{arctanh}(0) = 0$.

3.2. Titration of a strong base with a strong acid

Here the procedure to derive an equation for the concentration of OH^- is equivalent to the procedure used to derive the corresponding equation for the concentration of H_3O^+ in the case of a strong acid as the analyte, and can be obtained immediately from Equation (12) by making the replacements $[H_3O^+] \leftrightarrow [OH^-]$, $V_a \leftrightarrow V_b$ and $M_a \leftrightarrow M_b$. In this way one obtains

$$[OH^-]^2 - \frac{M_b V_b - M_a V_a}{V_a + V_b} [OH^-] - K_w = 0 \quad (21)$$

Using the same replacements and Equations (15)–(17), we can write immediately an expression for the pOH as a function of the volume V_a of titrant. The expressions for the pH of the solution being titrated are then given by

$$pH = pK_w - pOH \quad (22)$$

$$= \frac{1}{2}pK_w + \frac{\operatorname{arcsinh}\left(\frac{1}{2\sqrt{K_w}} \frac{M_b V_b - M_a V_a}{V_a + V_b}\right)}{\ln(10)} \quad (23)$$

$$= \frac{1}{2}pK_w + \frac{\operatorname{arccosh}\left(\sqrt{\frac{1}{4K_w} \left(\frac{M_b V_b - M_a V_a}{V_a + V_b}\right)^2 + 1}\right)}{\ln(10)} \quad (24)$$

$$= \frac{1}{2}pK_w + \frac{\operatorname{arctanh}\left(\frac{\frac{M_b V_b - M_a V_a}{V_a + V_b}}{\sqrt{\left(\frac{M_b V_b - M_a V_a}{V_a + V_b}\right)^2 + 4K_w}}\right)}{\ln(10)} \quad (25)$$

Note the presence of a plus sign before the inverse hyperbolic functions in Equations (23)–(25). This sign change produces a titration curve inverse to that of a strong acid with a strong base.

Figure 2(a) shows the titration curve obtained using any of Equations (23)–(25) for the case of the titration of 50 ml of a 0.1000 M strong base with 0.1000 M strong acid. Also shown in Figures 2(b) and 2(c) are the first and second derivatives of pH versus V_a obtained with Equations (18) and (19), making the corresponding replacements as indicated before. As in the case of the titration of a strong acid with a strong base, here also Equations (23)–(25) yield $pH = 7$ at the equivalence point.

4. TITRATION OF A WEAK MONOPROTIC ACID WITH A STRONG BASE

We now consider the case of the titration of a volume V_a of a

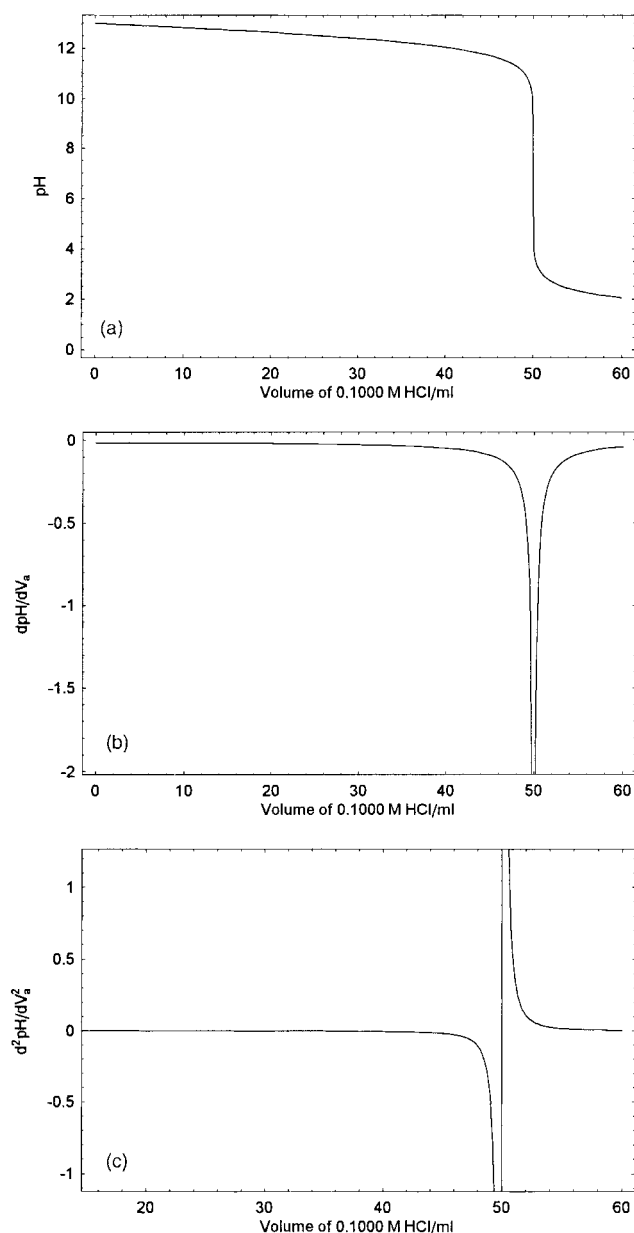


Figure 2. Titration of 50.00 ml of 0.1000 M NaOH with 0.1000 M HCl: (a) pH vs. V_a ; (b) $\text{dpH}/\text{d}V_a$; (c) $\text{d}^2\text{pH}/\text{d}V_a^2$.

weak monoprotic acid with equilibrium constant K_a and molar concentration M_a with a strong base of molar concentration M_b . The equation which allows the determination of $[\text{H}_3\text{O}^+]$ is a third-degree polynomial equation. This is an equation whose solution is not only feared by students but also by most textbook authors. The equation, which can be obtained by the principles of mass and charge balance [5], is given by

$$x^3 + ax^2 + bx - c = 0 \quad (26)$$

$$x = [\text{H}_3\text{O}^+] \quad (27)$$

$$a = \frac{M_b V_b}{V_b + V_a} + K_a \quad (28)$$

$$b = K_a \frac{M_b V_b - M_a V_a}{V_b + V_a} - K_w \quad (29)$$

$$c = K_a K_w \quad (30)$$

This equation possesses a single positive root, as can be determined by the application of Descartes' rule of signs [6]. This solution is given by

$$x = -\frac{1}{3}\xi_1 - \frac{2^{1/3}\xi_2}{3(\xi_3 + \sqrt{4\xi_2^3 + \xi_3^2})^{1/3}} + \frac{1}{32^{1/3}}\left(\xi_3 + \sqrt{4\xi_2^3 + \xi_3^2}\right)^{1/3} \quad (31)$$

where

$$\xi_1 = a \quad (32)$$

$$\xi_2 = 3b - a^2 \quad (33)$$

$$\xi_3 = 27c + 9ab - 2a^3 \quad (34)$$

Using this equation, we can obtain the pH by taking logarithms as

$$\text{pH} = -\log(x) \quad (35)$$

Unfortunately, in this case, owing to the presence of the cubic root, it is not possible to relate in an exact way the logarithm of the resulting expression with an inverse hyperbolic or trigonometric function. However, the titration curve and its derivatives can still be obtained by plotting directly Equation (35) with the definitions given by Equations (27–34). This is done in Figures 3(a)–3(c) for the case of the neutralization of 50.00 ml of 0.1000 M acetic acid ($K_a = 1.76 \times 10^{-5}$ M) with 0.1000 M NaOH. If in Equation (35) we let $V_b = 50.00$ ml, it is found that at the equivalence point $\text{pH} = 8.73$. It is important to note that even though in this case we cannot express the pH in terms of inverse hyperbolic functions of the volume of titrant, the shape of the titration curve is in general similar to that of those functions. The reason for this is that Equation (26) becomes Equation (12) when $K_a \rightarrow \infty$. Thus we can think of Figures 3(a)–3(c) as perturbed Figures 1(a)–1(c).

For weak and moderately weak acids it is a good approximation to neglect the cubic term in Equation (26), which, after rearrangement, becomes

$$x^2 + \left(\frac{K_a(V_b - M_a V_a/M_b) - K_w(V_a + V_b)/M_b}{V_b + K_a(V_a + V_b)/M_b} \right)x - \frac{K_a K_w(V_a + V_b)}{M_b[V_b + K_a(V_a + V_b)/M_b]} = 0$$

Comparing Equation (36) with Equation (4) and using Equation (9) allows us to write the pH in this case as

$$\text{pH} = -\text{p}\beta - \frac{1}{\ln(10)} \text{arcsinh}(\xi) \quad (37)$$

where ξ is given by Equation (8) with

$$b = \frac{K_a(V_b - M_a V_a/M_b) - K_w(V_a + V_b)/M_b}{V_b + K_a(V_a + V_b)/M_b} \quad (38)$$

$$c = -\frac{K_a K_w(V_a + V_b)}{M_b[V_b + K_a(V_a + V_b)/M_b]} \quad (39)$$

and β is chosen, as stated before, as the value that makes

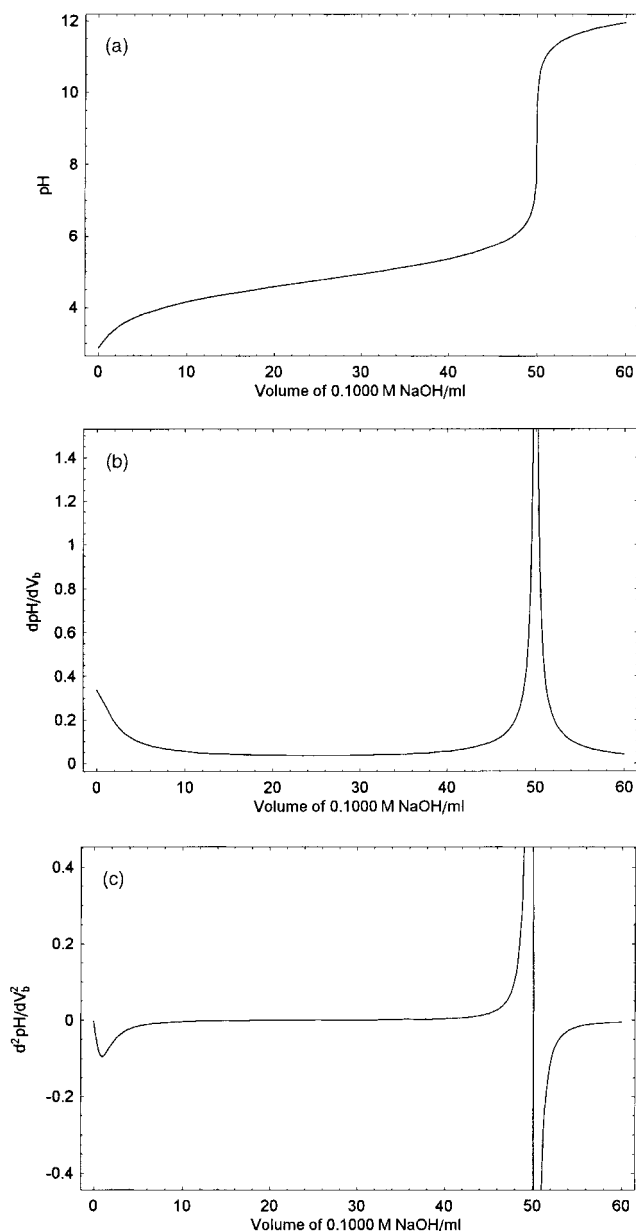


Figure 3. Titration of 50.00 ml of 0.1000 M acetic acid with 0.1000 M NaOH: (a) pH vs V_b ; (b) $\text{dpH}/\text{d}V_b$; (c) $\text{d}^2\text{pH}/\text{d}V_b^2$.

$\xi = 0$ at the equivalence point:

$$\beta = \left[\frac{K_w^2 (V_a + V_{eq})^2}{2M_b^2 [V_{eq} + K_a(V_a + V_{eq})/M_b]^2} \times \left(1 + \sqrt{1 + 4 \frac{K_a M_b [V_{eq} + K_a(V_a + V_{eq})/M_b]}{K_w (V_a + V_{eq})}} \right) + \frac{K_a K_w (V_a + V_{eq})}{M_b [V_{eq} + K_a(V_a + V_{eq})/M_b]} \right]^{-1/2} \quad (40)$$

A simplification of ξ is obtained by recognizing that, because $b > c$, Equation (8) can be written approximately as

$$\begin{aligned} \xi &= \frac{b}{2c\beta} \\ &= \frac{1}{2\beta K_w} \left(\frac{(V_{eq} - V_b)M_b}{V_a + V_b} + \frac{K_w}{K_a} \right) \end{aligned} \quad (41)$$

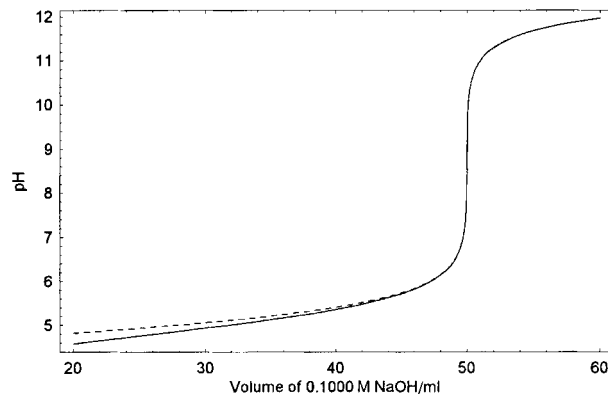


Figure 4. Titration of 50.00 ml of 0.1000 M acetic acid with 0.1000 M NaOH. Exact curve, Equation (35) and approximated, Equation (37) (both indistinguishable, full line), and approximated, Equation (42) (broken line).

Thus Equation (37) can be written approximately as

$$\text{pH} = -\text{p}\beta - \frac{1}{\ln(10)} \text{arcsinh} \left(\frac{1}{2\beta K_w} \left(\frac{(V_{eq} - V_b)M_b}{V_a + V_b} + \frac{K_w}{K_a} \right) \right) \quad (42)$$

and the other two equivalent forms in terms of the arccosh and arctanh functions. Note that, if we let $K_a \rightarrow \infty$ in Equations (40) and (42), we obtain Equation (15).

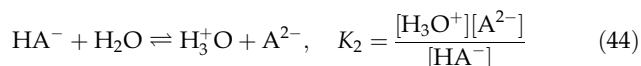
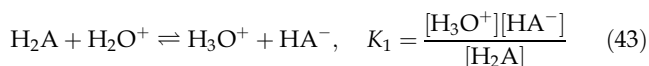
Figure 4 shows plots of Equations (37) and (42) for the neutralization of 50.00 ml of 0.1000 M acetic acid ($K_a = 1.76 \times 10^{-5}$ M) with 0.1000 M NaOH. For comparison, the exact result, obtained by solving the cubic equation, is also shown. Thus Equation (37) reproduces very accurately almost the entire titration curve, while Equation (42) is a good approximation which can be used for fitting experimental data.

5. TITRATION OF A WEAK MONOPROTIC BASE WITH A STRONG ACID

The case of the titration of a weak base with a strong acid can be treated similarly by making in Equations (26)–(35) the corresponding replacements indicated above and additionally $K_a \leftrightarrow K_b$. The curve obtained will be the inverse of the one corresponding to the titration of a weak acid with a strong base; this is shown in Figures 5(a)–5(c).

6. TITRATION OF DIPROTIC ACIDS AND BASES

We now consider the titration of a volume V_a of a diprotic acid H_2A with equilibrium constants K_1 and K_2 and molar concentration M_a with a volume V_b of a strong base BOH of molar concentration M_b . The corresponding equilibria are given by



The exact polynomial equation, which determines the concentration of H_3O^+ at each point of the titration process, can be obtained easily from the mass and charge conservation equations

$$\frac{M_a V_a}{V_b + V_a} = [\text{H}_2\text{A}] + [\text{HA}^-] + [\text{A}^{2-}] \quad (46)$$

$$[\text{H}_3\text{O}^+] + [\text{B}^+] = \frac{K_w}{[\text{H}_3\text{O}^+]} + [\text{HA}^-] + 2[\text{A}^{2-}] \quad (47)$$

along with the corresponding equilibrium constant Equations (43)–(45). The resulting polynomial equation is of fourth degree and is given by

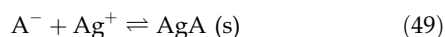
$$\begin{aligned} &[\text{H}_3\text{O}^+]^4 + \left(K_1 + \frac{M_b V_b}{V_b + V_a}\right)[\text{H}_3\text{O}^+]^3 \\ &- \left(K_w + K_1 \frac{M_a V_a - M_b V_b}{V_b + V_a} - K_1 K_2\right)[\text{H}_3\text{O}^+]^2 \\ &- \left(K_1 K_w + K_1 K_2 \frac{2M_a V_a - M_b V_b}{V_b + V_a}\right)[\text{H}_3\text{O}^+] - K_w K_1 K_2 = 0 \end{aligned} \quad (48)$$

The pH will then be given by $-\log[\text{H}_3\text{O}^+]$, where $[\text{H}_3\text{O}^+]$ is the positive root of Equation (48). Here the presence of a quartic root does not permit us to obtain a simple relation involving inverse hyperbolic or trigonometric functions. Figure 6 shows the plot of pH versus the volume of titrant for the titration of 50.00 ml of 0.1000 M maleic acid ($K_1 = 1.2 \times 10^{-2}$ and $K_2 = 6.0 \times 10^{-7}$) with 0.1000 M NaOH obtained by solving exactly Equation (48). The first- and second-stage equivalence point pHs have values of 4.12 and 9.37 respectively. It is important to note that, even though in the case of diprotic acids we cannot express the pH as a function of an inverse hyperbolic function, the general form of the titration curve looks like a superposition of perturbed titration curves of strong acids. The reason is that Equation (48) becomes Equation (12) when $K_1, K_2 \rightarrow \infty$. This observation forms the base of the mathematical modeling of titrations of polyprotic acids with strong bases as discussed below.

For the titration of a diprotic base with a strong acid, similar equations can be obtained by making the corresponding replacements stated above and additionally $K_{ai} \leftrightarrow K_{bi}$, $i = 2, 3, \dots$. The cases of titrations of mixtures of acids or bases can be treated in the same manner.

7. PRECIPITATION TITRATIONS

As a typical example we consider the titration of the halides Cl^- , Br^- and I^- with silver(I). Let A^- denote the halide, V_{Ag^+} and M_{Ag^+} the titrant volume and molar concentration respectively and V_{A^-} and M_{A^-} the corresponding quantities of the halide solution. Then the titration process can be represented by



with

$$K_{\text{sp}} = [\text{A}^-][\text{Ag}^+] \quad (50)$$

The concentration of the halide at each step of the titration

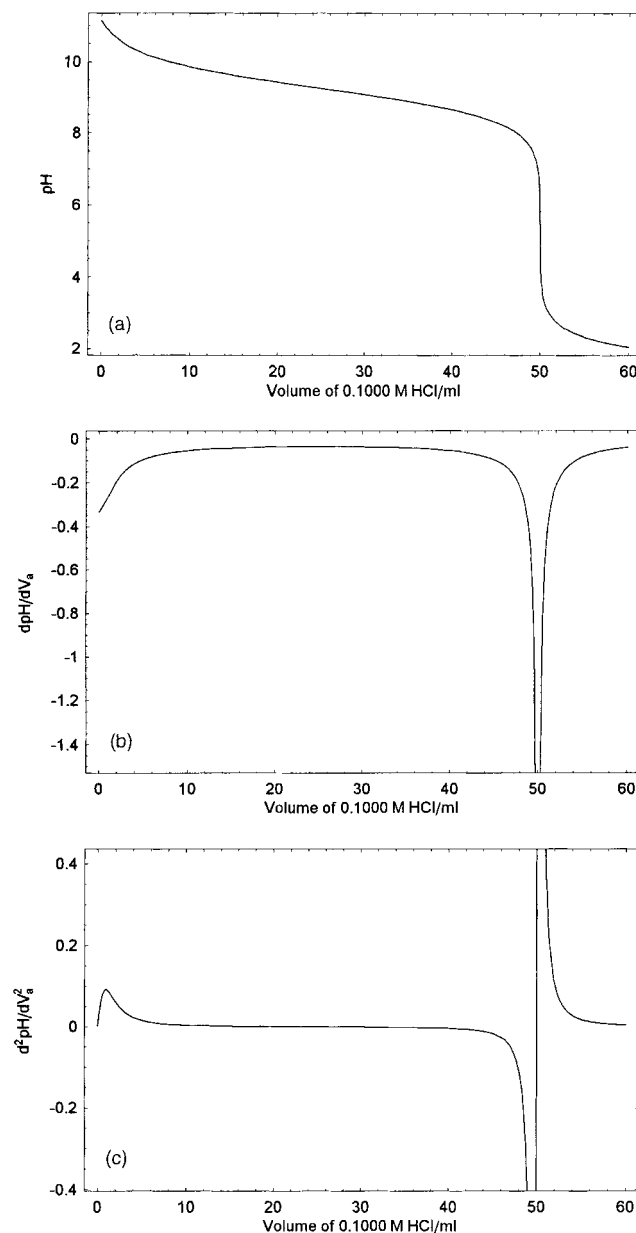


Figure 5. Titration of 50.00 ml of 0.1000 M NH_3 with 0.1000 M HCl: (a) pH vs V_a ; (b) $\text{dpH}/\text{d}V_a$; (c) $\text{d}^2\text{pH}/\text{d}V_a^2$.

process can be represented by

$$[\text{A}^-] = \frac{M_{\text{A}^-} V_{\text{A}^-} - M_{\text{Ag}^+} V_{\text{Ag}^+}}{V_{\text{A}^-} + V_{\text{Ag}^+}} + [\text{Ag}^+] \quad (51)$$

$$= \frac{M_{\text{A}^-} V_{\text{A}^-} - M_{\text{Ag}^+} V_{\text{Ag}^+}}{V_{\text{A}^-} + V_{\text{Ag}^+}} + \frac{K_{\text{sp}}}{[\text{A}^-]} \quad (52)$$

Thus the polynomial equation whose solution determines $[\text{A}^-]$ is given by

$$[\text{A}^-]^2 - \frac{M_{\text{A}^-} V_{\text{A}^-} - M_{\text{Ag}^+} V_{\text{Ag}^+}}{V_{\text{A}^-} + V_{\text{Ag}^+}} [\text{A}^-] - K_{\text{sp}} = 0 \quad (53)$$

This last equation has the same form as Equation (12). Thus

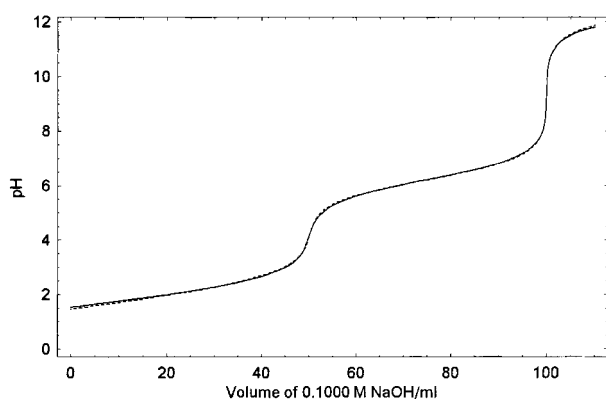


Figure 6. Titration of 50.00 ml of maleic acid with 0.1000 M NaOH. Exact curve, Equation (48) (full line) and fitted, Equation (106) (broken line).

we can write, using Equations (15)–(17),

$$\text{pA} = \frac{1}{2}\text{pK}_{\text{sp}} - \frac{\text{arcsinh}\left(\frac{1}{2\sqrt{K_{\text{sp}}}} \frac{M_{\text{A}^-}V_{\text{A}^-} - M_{\text{Ag}^+}V_{\text{Ag}^+}}{V_{\text{A}^-} + V_{\text{Ag}^+}}\right)}{\ln(10)} \quad (54)$$

$$= \frac{1}{2}\text{pK}_{\text{sp}} - \frac{\text{arccosh}\left(\sqrt{\frac{1}{4K_{\text{sp}}}\left(\frac{M_{\text{A}^-}V_{\text{A}^-} - M_{\text{Ag}^+}V_{\text{Ag}^+}}{V_{\text{A}^-} + V_{\text{Ag}^+}}\right)^2 + 1}\right)}{\ln(10)} \quad (55)$$

$$= \frac{1}{2}\text{pK}_{\text{sp}} - \frac{\text{arctanh}\left(\frac{\frac{M_{\text{A}^-}V_{\text{A}^-} - M_{\text{Ag}^+}V_{\text{Ag}^+}}{V_{\text{A}^-} + V_{\text{Ag}^+}}}{\sqrt{\left(\frac{M_{\text{A}^-}V_{\text{A}^-} - M_{\text{Ag}^+}V_{\text{Ag}^+}}{V_{\text{A}^-} + V_{\text{Ag}^+}}\right)^2 + 4K_{\text{sp}}}}\right)}{\ln(10)} \quad (56)$$

Figure 7 shows the curves of the titration of 50.00 ml of 0.1000 M NaCl or 50.00 ml of 0.1000 M NaBr with 0.1000 M AgNO₃ using any of Equations (54)–(56). At the equivalence point those equations reveal that $\text{pCl} = \frac{1}{2}\text{pK}_{\text{sp}} = \frac{12(-\log 1.56 \times 10^{-10})}{2} = 4.90$, i.e. $[\text{Cl}^-] = \sqrt{K_{\text{sp}}} = \sqrt{1.56 \times 10^{-10} \text{ M}^2} = 1.25 \times 10^{-5} \text{ M}$, and $\text{pBr} = \frac{1}{2}\text{pK}_{\text{sp}} = \frac{12(-\log 5.0 \times 10^{-13})}{2} = 5.65$, i.e. $[\text{Br}^-] = \sqrt{K_{\text{sp}}} = \sqrt{5.0 \times 10^{-13} \text{ M}^2} = 7.07 \times 10^{-7} \text{ M}$.

8. COMPLEXATION TITRATIONS

This kind of titration involves the formation of a soluble complex or co-ordination compound and is formally equivalent to those discussed before. Consider the simplest case of a ligand L which combines with a metal ion M in a 1:1 ratio. The corresponding equilibrium is given by



with formation constant

$$K_{\text{f}} = \frac{[\text{ML}]}{[\text{M}][\text{L}]} \quad (58)$$

Consider now the titration of a volume V_{m} of a solution of

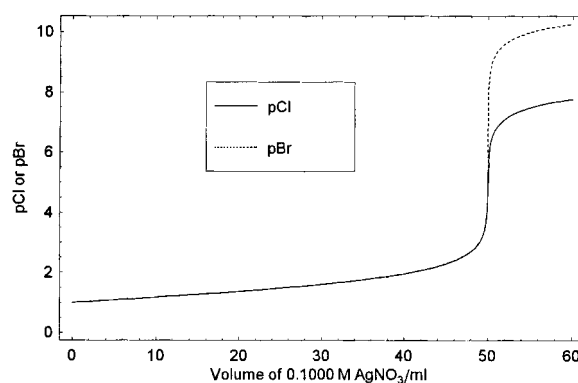


Figure 7. Titration of 50.00 ml of 0.1000 M NaCl or NaBr with 0.1000 M AgNO₃.

a metal ion of molar concentration M_{m} with a volume V_{l} of a ligand solution of molar concentration M_{l} . The corresponding polynomial equation for the concentration of the metal as a function of the volume of titrant can be obtained by combining Equation (58) with the mass balance equations [5]. It is given by

$$\left(\frac{V_{\text{l}}}{V_{\text{m}}} + 1\right)K_{\text{f}}[\text{M}]^2 + \left(\frac{V_{\text{l}}}{V_{\text{m}}} - K_{\text{f}}M_{\text{m}} + \frac{V_{\text{l}}}{V_{\text{m}}}K_{\text{f}}M_{\text{l}} + 1\right)[\text{M}] - M_{\text{m}} = 0 \quad (59)$$

Identifying b and c as

$$b = \frac{\frac{V_{\text{l}}}{V_{\text{m}}} - K_{\text{f}}M_{\text{m}} + \frac{V_{\text{l}}}{V_{\text{m}}}K_{\text{f}}M_{\text{l}} + 1}{\left(\frac{V_{\text{l}}}{V_{\text{m}}} + 1\right)K_{\text{f}}} \quad (60)$$

$$c = -\frac{M_{\text{m}}}{\left(\frac{V_{\text{l}}}{V_{\text{m}}} + 1\right)K_{\text{f}}} \quad (61)$$

in Equation (8), we can write pM in terms of inverse hyperbolic functions of the titrant volume, using Equations (9)–(11), as

$$\text{pM} = -\text{p}\beta - \frac{1}{\ln(10)} \text{arcsinh}(\xi) \quad (62)$$

with

$$\beta = \frac{1}{\sqrt{\frac{1}{4K_{\text{f}}^2} + \frac{M_{\text{m}}}{\left(1 + \frac{M_{\text{m}}}{M_{\text{l}}}\right)K_{\text{f}}} - \frac{1}{2K_{\text{f}}}}} \quad (63)$$

Since $b > c$ here, we can also use the approximation

$$\xi \approx \frac{b}{2c\beta} \quad (64)$$

$$= \frac{1}{2} \frac{K_{\text{f}}}{\beta} \left[\left(1 - \frac{V_{\text{l}}M_{\text{l}}}{V_{\text{m}}M_{\text{m}}}\right) - \frac{1}{K_{\text{f}}M_{\text{m}}} \left(1 + \frac{V_{\text{l}}}{V_{\text{m}}}\right) \right] \quad (65)$$

$$\approx \frac{1}{2} \frac{K_{\text{f}}}{\beta} \left(1 - \frac{V_{\text{l}}M_{\text{l}}}{V_{\text{m}}M_{\text{m}}}\right) \quad (66)$$

$$= \frac{1}{2} \frac{K_{\text{f}}}{\beta} \left(1 - \frac{V_{\text{l}}}{V_{\text{eq}}}\right) \quad (67)$$

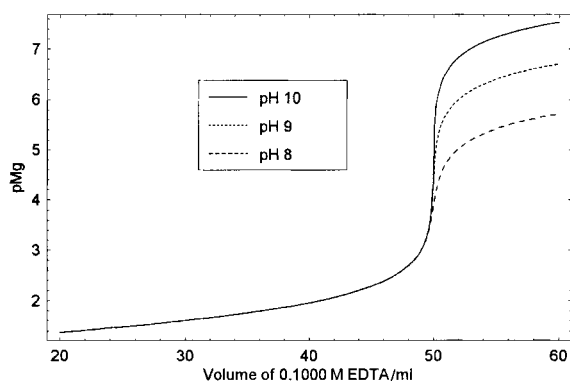
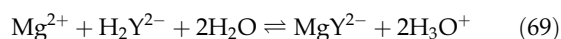


Figure 8. Titration of 50.00 ml of 0.1000 M Mg^{2+} with 0.1000 M EDTA at several pHs.

Thus a good approximation to pM is given by

$$\text{pM} = -\text{p}\beta - \frac{1}{\ln(10)} \operatorname{arcsinh} \left(\frac{1}{2} \frac{K_f}{\beta} \left(1 - \frac{V_1}{V_{\text{eq}}} \right) \right) \quad (68)$$

As a particular case, consider the titration of 50.00 ml of 0.1000 M Mg^{2+} with 0.1000 M EDTA at different pHs. In this case the reaction can be represented by



Here the formation constants in the equations must be substituted by *conditional formation constants* given by $K_f \propto \alpha$, where α is the fraction of the total uncomplexed EDTA that is in the form of Y^{4-} . We use the values $K_f = 4.9 \times 10^8$ and $\alpha = 5.4 \times 10^{-3}$, 5.2×10^{-2} and 3.5×10^{-1} at pH 8, 9 and 10 respectively [7]. Figure 8 shows the plot of Equation (62) for that titration at different pHs. At the equivalence point these equations indicate that $\text{pMg} = 3.86$, 4.35 and 4.76 (i.e. $[\text{Mg}^{2+}] = 1.38 \times 10^{-4}$, 4.47×10^{-5} and 1.74×10^{-5} M) at pH 8, 9 and 10 respectively. Figure 9 shows the comparison of the approximate Equation (68) with the exact Equation (62). As seen, Equation (68) gives a very accurate representation of the titration curve for almost the entire range of interest and, as such, is appropriate for fitting titration data for complexometric titration curves of the kind considered in this section.

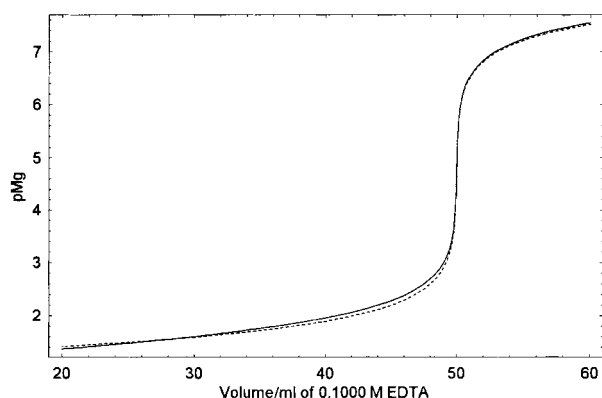
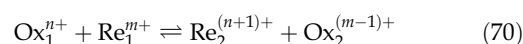


Figure 9. Titration of 50.00 ml of 0.1000 M Mg^{2+} with 0.1000 M EDTA. Exact curve, Equation (62) (full line) and approximated, Equation (68) (broken line).

9. REDOX TITRATIONS

Redox titrations deal with reactions where an electronic transfer between reactant and titrant occurs. Also, the two half-reactions of any redox titration system are always in equilibrium at all points after the beginning of the titration, so the half-cell reduction potentials are equal for all points of the titration process. In this way, changes in the cell potential E_{cell} can be used to follow the titration process and obtain a typical titration curve by plotting E_{cell} versus volume of titrant.

Here we consider only the case where an equimolar relationship between oxidant and reductor exists. For this case the general redox equation can be written as



Let V_o be the volume of a solution containing Ox_1^{n+} ions at a concentration M_o being titrated with a volume V_r of a solution containing Re_1^{m+} ions at a concentration M_r . For this reaction we have the equilibrium relation

$$K_e = \frac{[\text{Re}_2^{(n+1)+}][\text{Ox}_2^{(m-1)+}]}{[\text{Ox}_1^{n+}][\text{Re}_1^{m+}]} \quad (71)$$

and the following expression for the half-cell potential:

$$E_{\text{cell}} = E_{\text{Re}_2^{(n+1)+}/\text{Ox}_1^{n+}}^0 + 0.059 \log \left(\frac{[\text{Re}_2^{(n+1)+}]}{[\text{Ox}_1^{n+}]} \right) \quad (72)$$

By mass conservation, $[\text{Re}_2^{(n+1)+}]$ and $[\text{Ox}_1^{n+}]$ are given by

$$[\text{Re}_2^{(n+1)+}] = \frac{V_r M_r}{V_o + V_r} - [\text{Re}_1^{m+}] \quad (73)$$

$$= \frac{V_r M_r}{V_o + V_r} - \frac{[\text{Re}_2^{(n+1)+}][\text{Ox}_2^{(m-1)+}]}{[\text{Ox}_1^{n+}] K_e} \quad (74)$$

$$= \frac{V_r M_r}{V_o + V_r} - \frac{[\text{Re}_2^{(n+1)+}]^2}{[\text{Ox}_1^{n+}] K_e} \quad (75)$$

and

$$[\text{Ox}_1^{n+}] = \frac{V_o M_o - V_r M_r}{V_o + V_r} + [\text{Re}_1^{m+}] \quad (76)$$

$$= \frac{V_o M_o - V_r M_r}{V_o + V_r} + \frac{[\text{Re}_2^{(n+1)+}][\text{Ox}_2^{(m-1)+}]}{[\text{Ox}_1^{n+}] K_e} \quad (77)$$

$$= \frac{V_o M_o - V_r M_r}{V_o + V_r} + \frac{[\text{Re}_2^{(n+1)+}]^2}{[\text{Ox}_1^{n+}] K_e} \quad (78)$$

If we let $y = [\text{Re}_2^{(n+1)+}]$ and $x = [\text{Ox}_1^{n+}]$, the preceding equations can be written as

$$y = \frac{V_r M_r}{V_o + V_r} - \frac{y^2}{K_e x} \quad (79)$$

$$x = \frac{V_o M_o - V_r M_r}{V_o + V_r} + \frac{y^2}{K_e x} \quad (80)$$

Solving for y and x , we obtain

$$y = \frac{K_e (M_o V_o + M_r V_r)}{2(K_e - 1)(V_o + V_r)} - \frac{1}{2(K_e - 1)} \times \left(\frac{K_e^2 (M_o V_o + M_r V_r)^2}{(V_o + V_r)^2} - 4 \frac{(K_e - 1) K_e M_o M_r V_o V_r}{(V_o + V_r)^2} \right)^{1/2} \quad (81)$$

and

$$x = \frac{y^2}{K_e \left(\frac{V_r M_r}{V_o + V_r} - y \right)} \quad (82)$$

E_{cell} will be given by

$$E_{\text{cell}} = E_{\text{Re}_2^{(n+1)+}/\text{Ox}_1^{n+}}^0 + 0.059 \log \left(\frac{y}{x} \right) \quad (83)$$

Substituting Equations (81) and (82) in Equation (83), and after some cancellation, we obtain

$$E_{\text{cell}} = E_{\text{Re}_2^{(n+1)+}/\text{Ox}_1^{n+}}^0 + \frac{0.059}{2 \ln(10)} \ln(K_e) + \frac{0.059}{\ln(10)} \ln \left(\frac{\frac{2V(1-1/K_e)M_r V_r}{(M_o V_o + M_r V_r)} - 1}{1 - \sqrt{1 - 4 \frac{(1-1/K_e)M_o M_r V_o V_r}{(M_o V_o + M_r V_r)^2}}} \right) \sqrt{K_e} \quad (84)$$

It is not apparent that an $\text{arcsinh}(z)$ function is contained in Equation (84). To show that indeed it is there, consider the function of V contained in large parentheses in Equation (84), i.e.

$$\left(\frac{\frac{2V(1-1/K_e)M_r V_r}{(M_o V_o + M_r V_r)} - 1}{1 - \sqrt{1 - 4 \frac{(1-1/K_e)M_o M_r V_o V_r}{(M_o V_o + M_r V_r)^2}}} \right) \sqrt{K_e} \quad (85)$$

and let

$$a = \frac{2(1-1/k)M_r V_r}{(M_o V_o + M_r V_r)} \quad (86)$$

$$b = 4 \frac{(1-1/K_e)M_o M_r V_o V_r}{(M_o V_o + M_r V_r)^2} \quad (87)$$

With these definitions the expression (85) can be written as

$$\left(\frac{a}{1 - \sqrt{1-b}} - 1 \right) \sqrt{K_e} \quad (88)$$

Now we equate this last term to the function $\xi + \sqrt{1 + \xi^2}$, i.e.

$$\left(\frac{a}{1 - \sqrt{1-b}} - 1 \right) \sqrt{K_e} = \xi + \sqrt{1 + \xi^2} \quad (89)$$

and solve for ξ to get

$$\xi = \frac{K_e(\alpha - 1)^2 - 1}{2\sqrt{K_e}(\alpha - 1)} \quad (90)$$

$$\alpha = \frac{a}{1 - \sqrt{1-b}} \quad (91)$$

Using Equations (86)–(90) and (1), Equation (84) can now be

rewritten as

$$E_{\text{cell}} = E_{\text{Re}_2^{(n+1)+}/\text{Ox}_1^{n+}}^0 + \frac{0.059}{2 \ln(10)} \ln(K_e) + \frac{0.059}{\ln(10)} \ln \left(\xi + \sqrt{1 + \xi^2} \right) \quad (92)$$

$$= E_{\text{Re}_2^{(n+1)+}/\text{Ox}_1^{n+}}^0 + \frac{0.059}{2} \log(K_e) + \frac{0.059}{\ln(10)} \text{arcsinh}(\xi) \quad (93)$$

$$= E_{\text{Re}_2^{(n+1)+}/\text{Ox}_1^{n+}}^0 + \frac{0.059}{2} \log(K_e) + \frac{0.059}{\ln(10)} \text{arcsinh} \left(\frac{K_e(\alpha - 1)^2 - 1}{2\sqrt{K_e}(\alpha - 1)} \right) \quad (94)$$

with α given by Equation (91), and equivalent equations can be written in terms of arccosh and arctanh functions using Equations (2) and (3).

At the equivalence point we let $M_r V_r = M_o V_o$ in Equation (84) or (94) and find

$$E_{\text{cell}} = E_{\text{Re}_2^{(n+1)+}/\text{Ox}_1^{n+}}^0 + \frac{0.059}{2} \log(K_e) + \frac{0.059}{\ln(10)} \ln(1) \quad (95)$$

$$= E_{\text{Re}_2^{(n+1)+}/\text{Ox}_1^{n+}}^0 + \frac{0.059}{2} \log(K_e) + \frac{0.059}{\ln(10)} \text{arcsinh}(0) \quad (96)$$

$$= E_{\text{Re}_2^{(n+1)+}/\text{Ox}_1^{n+}}^0 + \frac{0.059}{2 \ln(10)} \ln(K_e) \quad (97)$$

A good approximation to Equation (94) can be found. Near the equivalence point, and considering that K_e is a large number, we can approximate $\alpha - 1$ as

$$\alpha - 1 \approx \frac{\frac{2(1-1/k)M_r V_r}{M_o V_o + M_r V_r}}{1 - \sqrt{\frac{4}{K_e} \frac{M_o M_r V_o V_r}{(M_o V_o + M_r V_r)^2}}} \quad (98)$$

$$\approx \frac{M_r V_r - M_o V_o}{M_r V_r + M_o V_o} \quad (99)$$

Also, ξ in Equation (90) can be written as

$$\xi \approx \sqrt{K_e}(\alpha - 1) \quad (100)$$

$$= \sqrt{K_e} \frac{M_r V_r - M_o V_o}{M_r V_r + M_o V_o} \quad (101)$$

Thus, using Equation (101), an approximation to Equation (94) is given by

$$E_{\text{cell}} = E_{\text{Re}_2^{(n+1)+}/\text{Ox}_1^{n+}}^0 + \frac{0.059}{2} \log(K_e) - \frac{0.059}{\ln(10)} \text{arcsinh} \left(\sqrt{K_e} \frac{M_o V_o - M_r V_r}{M_r V_r + M_o V_o} \right) \quad (102)$$

Consider the titration of 50.00 ml of 0.1000 M Fe^{2+} with 0.1000 M Ce^{4+} in a medium 0.5 M with respect to H_2SO_4 . For this reaction the equilibrium constant has the value $K_e = 7.9 \times 10^{12}$ and $E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^0 = 0.700$ V. The value of E_{cell} at the equivalence point is calculated from Equation (97) as 1.080 V.

Figure 10 shows the titration curve for E_{cell} versus V_r using Equation (94). This equation, expressed in terms of an arcsinh function, reproduces the known titration curve for this reaction. Also plotted in the same figure is Equation (102). As seen, the approximate equation reproduces with

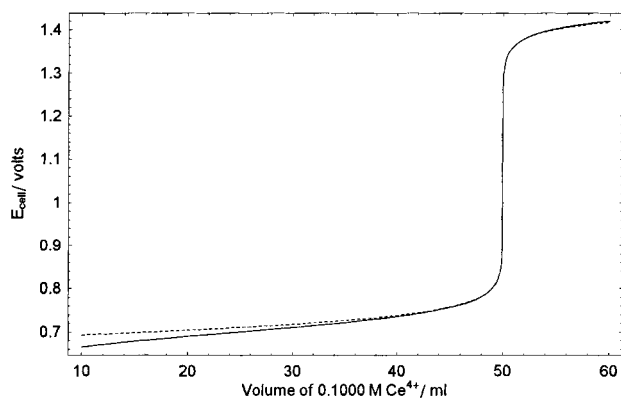


Figure 10. Titration of 50.00 ml of 0.1000 M Fe^{2+} with 0.1000 M Ce^{4+} in 0.5 M H_2SO_4 . Exact curve, Equation (94) (full line) and approximated, Equation (102) (broken line).

accuracy the exact titration curve and would be useful for fitting experimental data for this kind of reaction, or even for more complicated ones, as will be shown in the next section.

10. APPROXIMATE MODELING OF TITRATION CURVES

10.1. Modeling of sigmoidal titration curves

Previously we studied the simplest case of several types of titrations and saw that the presence of a square root in the expression of $-\log(M)$ as a function of titrant volume allows us to re-express the relation in terms of inverse hyperbolic functions. In the case of more complicated titrations such as the neutralization of a weak acid by a strong base and other types of titrations where the stoichiometric relation between reactant and titrant is not 1:1, the presence of cubic root and higher-order equations does not allow us to obtain a simple expression relating $-\log(M)$ with the volume of titrant in terms of inverse hyperbolic functions. However, as shown graphically above, it is still possible to construct good *approximate* models for the entire titration curves. We have shown in the preceding sections (see Equations (15), (23), (42), (54), (68) and (102)) that, in general, the dependence of the titration variable f (where f could be pM , M being the concentration of the species titrated, or E_{cell}) on the titrant volume V_t will be given by a relation of the type

$$f(V_t) = \log(\beta) - \frac{b}{\ln(10)} \operatorname{arcsinh}\left(c \frac{d - V_t}{e + gV_t}\right) \quad (103)$$

where β , b , c , d , e and g are constants independent of the titrant volume V_t .

Equation (103) yields the fundamental parameters of any titration: $\log(\beta)$, the value of $f(V_t)$ at the equivalence point, and d , the equivalence point volume.

Taking as an example the titration of 50.00 ml of 0.1000 M acetic acid with 0.1000 M NaOH (see e.g. Table 6.5 of Reference [8]) and fitting the data to Equation (103), we obtain $\text{pH} = 8.73$ and $V_t = 50.00$ ml at the equivalence point. The corresponding titration curve is shown in Figure 11 and, as observed, is an accurate representation of the exact curve, which is shown for comparison.

In the case of redox reactions we can also use an

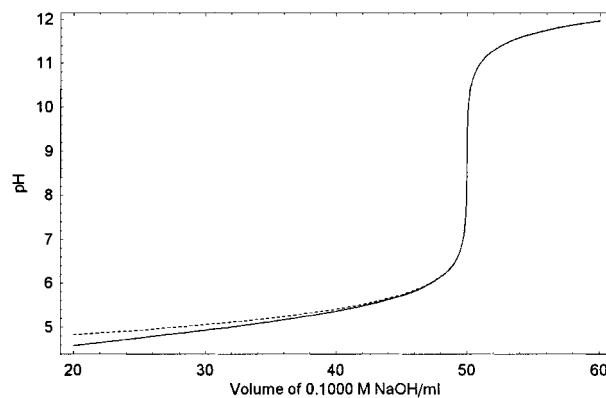
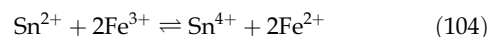


Figure 11. Titration of 50.00 ml of acetic acid with 0.1000 M NaOH. Fitted curve according to Equation (103) (broken line) and exact, Equation (35) (full line).

expression like Equation (103) to fit the titration data. If we use, for instance, the data, volume of Ce^{4+} vs $[\text{Fe}^{2+}]$, given in Table 9.1 of Reference [8] for the titration of 50.00 ml of 0.1000 M Fe^{2+} with 0.1000 M Ce^{4+} in 0.5 M H_2SO_4 and fit them to an equation of the form of Equation (103), we find the parameters $E_{\text{cell}} = 1.08$ V and $V_t = 50.00$ ml at the equivalence point. Figure 12 shows a plot of the resulting fitted function together with the exact curve for comparison.

The last two examples were chosen because we know from the analysis developed in the preceding sections that the fitting to Equation (103) has to be accurate. As an example of a titration reaction which corresponds to a non-equimolar relationship, consider the titration of 50.00 ml of 0.0500 M Sn^{2+} with a solution of 0.1000 M Fe^{3+} in 2 M HCl. The titration reaction is given by



Fitting the data, volume of Fe^{3+} vs $[\text{Sn}^{2+}]$, reported in Table 9.2 of Reference [8] to Equation (103), we get $E_{\text{cell}} = 0.33$ V and $V_t = 50.00$ ml at the equivalence point, which are also the exact values. Figure 13 shows a plot of the resulting fitted equation together with the 'simulated points' calculated using the Nernst equation.

As a final example of another non-equimolar reaction,

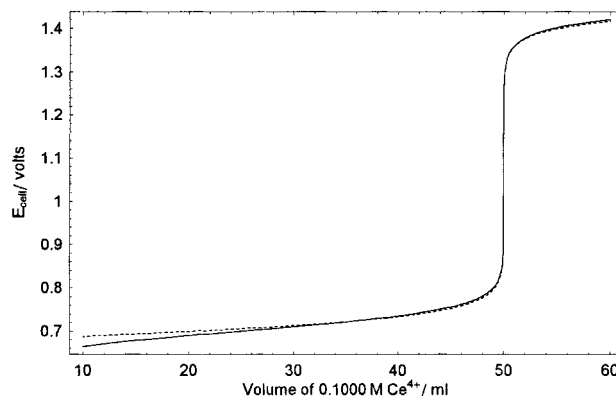


Figure 12. Titration of 50.00 ml of 0.1000 M Fe^{2+} with 0.1000 M Ce^{4+} in 0.5 M H_2SO_4 . Fitted curve according to Equation (103) (broken line) and exact, Equation (94).

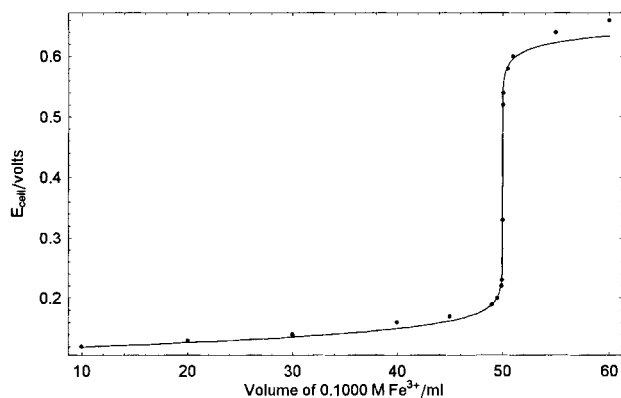
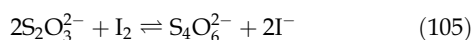


Figure 13. Titration of 50.00 ml of 0.0500 M Sn^{2+} with 0.1000 M Fe^{3+} in 2 M HCl. Simulated points and fitted equation of the form of Equation (103).

consider the case of the titration of 50.00 ml of 0.0500 M I_2 with 0.1000 M $\text{Na}_2\text{S}_2\text{O}_3$ at $\text{pH} \neq 6$ and 10^{-2} M I^- . The reaction is given by



The form of the titration curve of this reaction is inverse to those of the titrations considered up to now in this subsection, which means that the parameter b in Equation (103) is negative. For this case we fit the data, volume of $\text{Na}_2\text{S}_2\text{O}_3$ vs $[\text{I}_2]$, given in Table 9.3 of Reference [8] to Equation (103) and obtain $E_{\text{cell}} = 0.42$ V and $V_t = 50.00$ ml at the equivalence point. The plot of the resulting fitted function is shown in Figure 14 along with the 'simulated points' calculated using the Nernst equation. As seen, Equation (103) reproduces the inverse curve as well, since it simply changes the sign of the b parameter. Other cases that we have studied indicate that Equation (103) is a remarkably good model for redox titration curves.

Now, what about the case of diprotic acids? Could we construct a theoretical model to reproduce their titration curves? To construct our theoretical model based on the

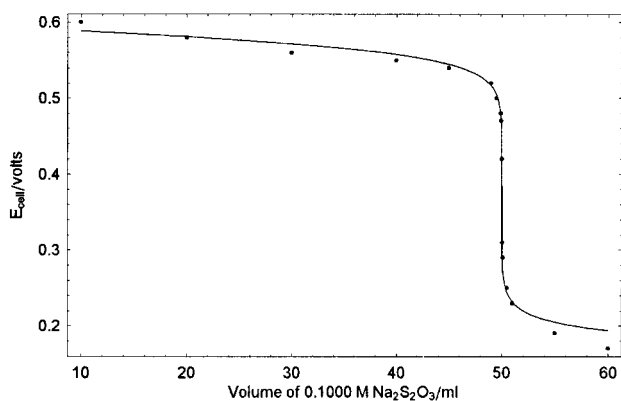


Figure 14. Titration of 50.00 ml of I_2 solution with 0.1000 M $\text{Na}_2\text{S}_2\text{O}_3$ at $\text{pH} 6.0$ and $[\text{I}^-] = 10^{-2}$ M. Simulated points and fitted equation of the form of Equation (103).

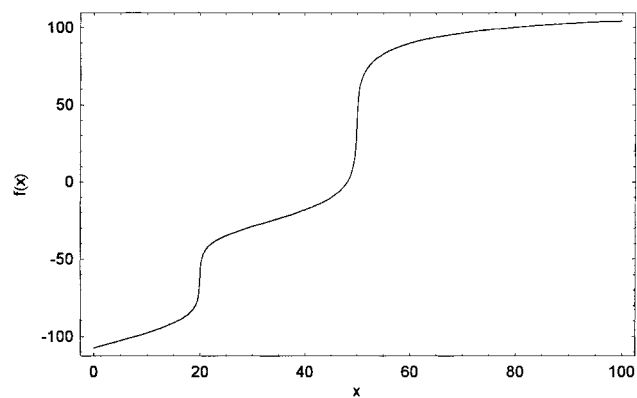


Figure 15. Plot of Equation (106) for selected values of the parameters.

derivations we have made in the last sections, observe the full curve in Figure 6. We can think of the titration curve of diprotic acids as two approximately independent curves, each one looking like the curve of an *arcsinh* function, as we have discussed before, such that as one of the curves starts to die the other is born. This observation leads us then to propose the following mathematical function to model the titration curve of a diprotic acid with a strong base:

$$\text{pH} = a_1 - b_1 \text{arcsinh} \left(c_1 \frac{d_1 - V}{d_1 + V} \right) - b_2 \text{arcsinh} \left(c_2 \frac{d_2 - V}{d_2 + V} \right) \quad (106)$$

As a test, consider the parameter values $a_1 = 5$, $b_1 = 5$, $c_1 = 700$, $d_1 = 20$, $b_2 = 10$, $c_2 = 1000$ and $d_2 = 50$. The graph of the resulting function is shown in Figure 15.

Thus let us now try Equation (106) as a mathematical model for the titration of 50.00 ml of 0.1000 M maleic acid with 0.1000 M NaOH using a set of data points generated by the exact solution of Equation (48). Figure 6 shows a plot of the fitted equation as well as the exact curve for comparison. As observed, the general features of the exact curve are well reproduced by the approximate one. Equation (106) predicts that the first- and second-stage equivalence point pHs are 4.05 and 9.40 (exact: 4.12 and 9.37) respectively.

10.2. Gran-type equation modeling of titration curves

Gran plots obtain the equivalence point by linear extrapolations based on linear formulae obtained from chemical balance and neglect of small quantities before the equivalence point [2,9,10]. Our methodology up to now obtains the equivalence point and other chemical parameters by fitting directly to equations which *approximately reproduce the entire titration curve, based on simplifications of the exact equations.*

In this subsection we will obtain linear or, at most, polynomial approximations to our equations which can be useful in modeling some titration cases. Some of the linearizations that will be obtained are already well known as Gran equations [2], but we would like to emphasize how they are obtained as approximations to the equations derived in the preceding sections.

From the mathematical analysis of the titration equations

for the different types of titrations, we have written Equation (103) as the general, approximate representation of the titration equation. Before and close to the equivalence point the value of the argument of the arcsinh function is a large number; thus we can use the approximation $\text{arcsinh}(x) \approx \ln(2x)$ for $x^2 > 1$ [3] in Equation (103). In this way, and after some rearrangement, Equation (103) can be rewritten approximately as

$$(e + gV_t)10^{-f(V_t)/b} = \frac{2c(d - V_t)}{\beta^{1/b}} \quad (107)$$

This equation will be the basis for our derivation of approximately linear relations for the different types of titrations discussed in the preceding sections.

In the first place, consider the case of the titration of a strong acid with a strong base. Our exact equation, which gives the pH as a function of V_{eq} , is given by Equation (15). Making the corresponding identification of parameters between Equations (15) and (103), we find, from Equation (107),

$$[\text{H}_3\text{O}^+](V_a + V_b) = \frac{M_b}{\beta\sqrt{K_w}}(V_{\text{eq}} - V_b) \quad (108)$$

However, we know that in this case $\beta = 1/\sqrt{K_w}$. Thus Equation (108) becomes

$$[\text{H}_3\text{O}^+](V_a + V_b) = M_b(V_{\text{eq}} - V_b) \quad (109)$$

which is the corresponding Gran equation for the titration of a strong base with a strong acid.

Now consider the case of the titration of a monoprotic weak acid with a strong base. The approximate non-linear equation that describes the process is given by Equation (42). By making the corresponding identification between Equations (42) and (103), we can write Equation (107) as

$$[\text{H}_3\text{O}^+](V_a + V_b) = \frac{M_b}{\beta^2 K_w}(V_{\text{eq}} - V_b) \quad (110)$$

For weak acids β is given by Equation (40). We can identify two limits of β from this equation. First, when the substance being titrated is a strong acid, $K_a \rightarrow \infty$ and $\beta = 1/\sqrt{K_w}$. Second, when the substance being titrated is a moderately weak acid,

$$\beta^2 \approx \frac{M_b[V_{\text{eq}} + K_a(V_a + V_{\text{eq}})/M_b]}{K_a K_w(V_a + V_{\text{eq}})}$$

Since the Gran plot is drawn using points near the equivalence point, we can approximate $V_{\text{eq}} \approx V_b$ in the expression for β and substitute the resulting expression in Equation (110) to obtain

$$[\text{H}_3\text{O}^+]\left(V_b + K_a \frac{V_a + V_b}{M_b}\right) = K_a(V_{\text{eq}} - V_b) \quad (111)$$

The preceding equation contains the Gran equations for two extreme cases of monoprotic acid titrations with strong bases. When $K_a \rightarrow \infty$ (strong acids), we get

$$[\text{H}_3\text{O}^+](V_a + V_b) = M_b(V_{\text{eq}} - V_b)$$

and when $K_a \rightarrow 0$ (very weak acids), we obtain

$$[\text{H}_3\text{O}^+]V_b = K_a(V_{\text{eq}} - V_b) \quad (112)$$

Second, consider the case of precipitation titrations with equimolar relationships. Equation (54) gives the exact equation for this process. By comparison with Equation (103) and use of Equation (107), or simply by analogy to the case of titration of a strong acid with a strong base, we find the Gran-type equation

$$[\text{A}^-](V_{\text{A}^-} + V_{\text{Ag}^+}) = M_{\text{Ag}^+}(V_{\text{eq}} - V_{\text{Ag}^+}) \quad (113)$$

with $V_{\text{eq}} = M_{\text{A}^-}V_{\text{A}^-}/M_{\text{Ag}^+}$. Thus a plot of $[\text{A}^-](V_{\text{A}^-} + V_{\text{Ag}^+})$ vs V_{Ag^+} will be linear, with negative slope, from which we can obtain V_{eq} .

Next, consider the case of complexometric titrations with an equimolar metal-ligand relationship. Our approximate equation for this process is given by Equation (68). By comparison of this equation with Equation (103) and use of Equation (107) we get the Gran-type equation

$$[\text{M}] = \frac{K_f}{\beta^2} \left(1 - \frac{V_1}{V_{\text{eq}}}\right) \quad (114)$$

A plot of Equation (114) for the titration of 50.00 ml of 0.1000 M Mg^{2+} with 0.1000 M EDTA is shown in Figure 16. As seen, it shows some curvature. This is a consequence of the fact that, contrary to the other cases considered up to now, the parameter c in Equation (103) is not large enough to tame the factor $1 - V_1/V_{\text{eq}}$ as it tends to zero at the equivalence point. In this case we will have to invert completely, i.e. with no approximation, Equation (68) with the help of Equation (1). In this way we find

$$[\text{M}] = \frac{1}{\beta} \left[\frac{1}{2\beta} K_f \left(1 - \frac{V_1}{V_{\text{eq}}}\right) + \sqrt{1 + \frac{K_f^2}{4\beta^2} \left(1 - \frac{V_1}{V_{\text{eq}}}\right)^2} \right] \quad (115)$$

$$= a_0 + a_1 V_1 + a_2 V_1^2 + \dots \quad (116)$$

Equations (115) and (116) show that at the equivalence point, when $V_1 = V_{\text{eq}}$, $[\text{M}] = 1/\beta$. Since β is a large number on the order of $\sqrt{K_f}$ (see Equation (63)), we can, without loss of accuracy, determine V_{eq} by fitting the data of $[\text{M}]$ vs V_1 to a polynomial of certain degree and finding the value of V_1 which makes $[\text{M}] = 0$. Figure 16 also shows the plot of the corresponding fitted fourth-order polynomial equation to the data. Using this procedure, we find $V_{\text{eq}} = 50.00$ ml.

Finally, we consider the case of redox titrations. For an equimolar reaction we have found that the process is described approximately by Equation (102). By making a comparison between this equation and Equation (103) and further use of Equation (107), we find, after some rearrangement,

$$(1 + 2 \times 10^{\Delta E/0.059})V_r = 2 \times 10^{\Delta E/0.059}V_{\text{eq}} - V_{\text{eq}} \quad (117)$$

As an illustration of the use of Equation (117), consider the titration of 50.00 ml of 0.1000 M Fe^{2+} with 0.1000 M Ce^{4+} in a medium 0.5 M with respect to H_2SO_4 . We use data given in Table 9.1 of Reference [8] and fit them to Equation (117). The plot of the resulting linear equation is shown in Figure 17 and yields $V_{\text{eq}} = 50.00$ ml. We have tested Equation (117) with non-equimolar reactions and have found that for stoichiometric coefficients not very different from unity, e.g. the cases of the reactions indicated in Equations (104)

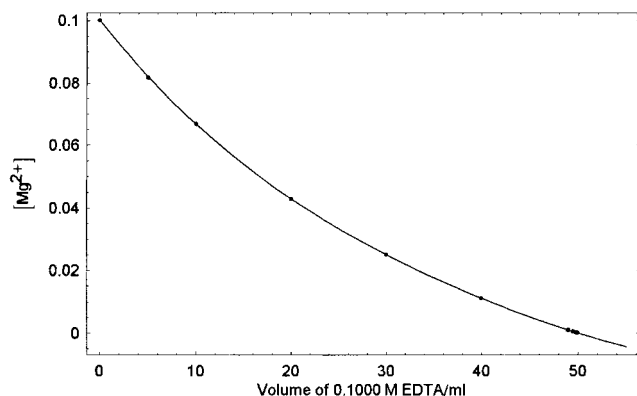


Figure 16. Titration of 50.00 ml of 0.1000 M Mg^{2+} with 0.1000 M EDTA. $[\text{Mg}^{2+}]$ vs V_{EDTA} ; simulated points and fitting to a fourth-degree polynomial.

and (105), Equation (117) reproduces the titration parameters with accuracy.

11. A SIMILARITY BETWEEN THE NMR SPECTRUM AND THE ACID-BASE NEUTRALIZATION PROCESS

An interesting coincidence that is worth considering is the fact that titration curves and those of integrated NMR signals have similar shapes. From this visual correspondence between the curves of titration and the integrated NMR signal it is inferred that the curve $\text{dpH}/\text{d}V$ versus V , V being the volume of titrant, must be similar in shape to the lineshape of an NMR signal versus frequency. That this true can be seen by comparing Figure 1(b) of the present paper with any ^1H NMR signal. This similarity can be understood if we consider that, as can be shown [11], the integrated curve of a ^1H NMR signal is expressed in terms of an $\arctan(x)$ function, as is the case also for the function $\text{pM}(V_t)$ or $E_{\text{cell}}(V_t)$ discussed previously. This similarity of the shapes of the curves associated with two processes based on physically different mechanisms suggests that a connec-

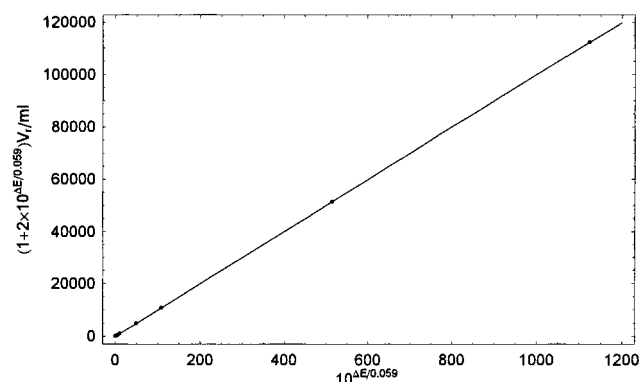


Figure 17. Titration of 50.00 ml of 0.1000 M Fe^{2+} with 0.1000 M Ce^{4+} in 0.5 M H_2SO_4 . Simulated data points and linear fitting to Equation (117).

tion between them must exist through an *equivalence relation between their variables and parameters*. Indeed, consider the titration of a strong acid with a strong base, both of the same concentration M , as a prototype example and establish the following correspondences:

$$\begin{aligned} \frac{\text{dpH}}{\text{d}V_b} &\Leftrightarrow I(\nu) \\ V_b &\Leftrightarrow \nu \\ V_a &\Leftrightarrow \nu_0 \\ T_2 &\Leftrightarrow \frac{M}{8V_a\sqrt{K_w}} \end{aligned} \quad (118)$$

where $I(\nu)$ is the NMR lineshape as a function of the frequency ν , T_2 is the transverse or spin-spin relaxation time and ν_0 is the resonance frequency. We can then say that the abrupt change in pH around the equivalence point, i.e. at $V_b \approx V_a$, is *equivalent* to the abrupt change in intensity that occurs around $\nu \approx \nu_0$. On the other hand, the spin-spin relaxation time is *equivalent*, basically, to the inverse of the square root of the ionic constant of water. Furthermore, both curves are concave down and have the same asymptotic behavior.

Thus we can say, from the similarity presented here, that *the absorption of a photon of frequency ν_0 by a nucleus of ^1H is equivalent to the neutralization of a volume V' of reactant by a titrant*.

12. CONCLUSIONS

In conclusion, we have shown mathematically the origin of the empirical observation that titration curves look remarkably similar to those of inverse hyperbolic or trigonometric functions, and have deduced the exact dependence of the argument of these functions on the volume of titrant and other chemical parameters, as well as accurate approximate non-linear equations for describing *the entire titration curve*; linearizations of those equations lead to Gran-type equations. Both types of equations, the non-linear and the linear ones, can be used to determine with accuracy the relevant parameters of titrations.

Some authors have tried to find general *implicit* relations which involve the volume of titrant as a function of the concentration of some of the species in the solution being titrated [5,12,13], or as a function of the cell potential in the case of potentiometric titrations [14]. However, this methodology sometimes leads to physically unrealizable values of the titrant volume. Here we have constructed *explicit* relations which express the negative logarithm of the concentration of some species in the solution being titrated as a function of the titrant volume. Because of this, no physically unrealizable situations can be encountered. Our results have led to *explicit generalized relations* which can be used in a number of simple reactions and as models in more complex ones.

Finally, we have shown that an interesting analogy exists between titration curves and integrated NMR signals, and have interpreted this in terms of a similarity between the equations that describe the two processes.

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